

Extra Formulas & Techniques Needed for the Exercises

(Not stated explicitly in the theory sections — organized by topic)

A. Matrix Algebra

A.1 Solving a matrix equation for an unknown matrix

Explanation: The theory defines the inverse ($A \cdot A^{-1} = I_n$) but never shows how to isolate an unknown matrix X from an equation like $X \cdot A = B$. The trick is the same as solving a scalar equation, but you must multiply **on the correct side** (matrix product is not commutative), and only if A is invertible. **Technique:** $X \cdot A = B \Rightarrow$ (right-multiply both sides by A^{-1}) $\Rightarrow X \cdot A \cdot A^{-1} = B \cdot A^{-1} \Rightarrow X = B \cdot A^{-1}$. (Similarly, $A \cdot X = B$ would require **left**-multiplying by A^{-1} : $X = A^{-1} \cdot B$.)

A.2 Matrix polynomials (A^2 , $A^2 - 2A$, etc.)

Explanation: The theory covers the product of two *different* matrices, but exercises ask you to compute powers/polynomials of a single matrix ($A^2 = A \cdot A$), then combine the result with scalar multiples of A using ordinary matrix addition/subtraction — a straightforward but non-obvious combination of two separate rules from the theory. **Example:** $A^2 - 2A = (A \cdot A) - 2 \cdot A$, computed entry-by-entry after first finding $A \cdot A$.

B. Determinants as Functions of a Parameter

B.1 Turning a determinant condition into a polynomial equation

Explanation: When a matrix depends on a parameter (a or k), "the determinant is zero" is not a single number to check but an **equation to solve**. You expand the determinant symbolically (it becomes a polynomial in the parameter), then find its roots — these roots are exactly the boundary cases that need separate discussion (singular matrix, dependent/inconsistent system, linear dependence, etc.). **Example:** $\det(A) = 2a - 2 + 4a^2 = 0$ has roots $a = 1/2$ and $a = -1$; the vectors are independent for every other value of a .

B.2 Factoring a parametric determinant

Explanation: Parametric determinants often factor nicely (e.g., into $(k-1)^2(k+2)$), which is not a new formula but ordinary polynomial factoring — necessary to read off the roots quickly instead of using the quadratic/cubic formula on an unfactored expression.

C. Rank and Linear (In)dependence

C.1 Using rank to test linear (in)dependence of vectors

Explanation: The theory states the *result* (k vectors are independent iff the rank of the matrix built from them equals k) but the practical skill is arranging vectors as rows or columns correctly and then computing that rank with minors — effectively re-using the "find the largest non-zero minor" procedure from Section 2.5, now applied to a question about vectors rather than an explicit system.

C.2 Reducing an over-determined system to a Cramer's system

Explanation: A consistent-dependent system with more equations than its rank r is not itself solvable directly by Cramer's rule (it isn't square). The technique: identify a non-zero $r \times r$ minor, discard the equations/columns not involved in it, treat the leftover variables as parameters, and solve the resulting $r \times r$ Cramer's system for each fixed value of the parameters. This combines the rank theorem (Section 2.5) with Cramer's rule (Section 4.2) in a way neither states outright.

D. Systems of Equations From Word Problems

D.1 Translating an equilibrium/balance condition into equations

Explanation: Several applied problems (network data flow, production/consumption models, sales data) never give you equations directly — you must translate a verbal condition ("what comes in equals what goes out," "total profit equals price times units summed over products") into one linear equation per node/year/constraint. This is an algebra-word-problem skill, not a theorem: identify the unknowns, then write one equation per given numerical fact or physical law.

D.2 The Leontief input-output model

Explanation: A production vector X , a consumption matrix C , and a demand vector D are related by $X = C \cdot X + D$. Solving for X requires recognizing X as $I \cdot X$, moving terms, and factoring: **Formula:** $X = C \cdot X + D \Rightarrow (I - C) \cdot X = D \Rightarrow X = (I - C)^{-1} \cdot D$ This is a direct but non-obvious combination of basic matrix algebra (Section 1.3) and the inverse-matrix formula (Section 2.4); it is not stated as a named formula in the theory, only demonstrated through the worked example.

E. Lines in the Plane

E.1 Converting between the different forms of a line's equation

Explanation: The theory derives all five forms (vectorial, parametric, continuous, slope-intercept, general) from a single point and direction vector, but exercises often give you a line already in one form and ask for a different one, or ask you to check whether a given point lies on the line — this requires running the derivation "in reverse" or substituting coordinates directly. **Example:** to check whether $Q(x, y, -3)$ lies on a line given in parametric form, substitute the known coordinate(s) into the parametric equations, solve for the parameter k , and use that same k to find the missing coordinate(s).

E.2 Cross-multiplication to obtain the general equation directly

Explanation: Going from the continuous equation $(x-p_1)/v_1 = (y-p_2)/v_2$ to the general equation $Ax+By+C=0$ uses the basic algebraic technique of cross-multiplying a proportion ($a/b = c/d \Rightarrow ad = bc$) and then collecting all terms on one side — ordinary algebra, not a new geometric fact, but the step is what produces the standard A, B, C coefficients ($A = v_2, B = -v_1, C = -v_2p_1 + v_1p_2$).

Quick-Reference Table

Situation	Formula / Technique	Why it's not explicit in the theory
Solve $X \cdot A = B$ for X	$X = B \cdot A^{-1}$ (multiply on the correct side)	Theory defines A^{-1} but not equation-solving with matrices
Solve $A \cdot X = B$ for X	$X = A^{-1} \cdot B$	Same as above, opposite side
Compute $A^2 - 2A$	$(A \cdot A) - 2A$, combining two known rules	Theory shows each rule separately, not combined
$\det(A)$ depends on parameter	Solve $\det(A)=0$ as a polynomial equation; factor it	Theory shows numeric determinants only
Test if vectors are independent	rank of matrix built from them, via minors	Theory states the rank-independence link as a fact, not a worked recipe
Over-determined dependent system	Drop extra equations, use non-zero $r \times r$ minor, treat rest as parameters, apply Cramer's rule	Combines rank theorem + Cramer's rule, not spelled out together
Leontief production model	$X = (I - C)^{-1}D$	Combines matrix algebra + inverse formula into a named economic model
Check a point lies on a line	Substitute into parametric/continuous form and solve for k	Theory derives the forms but not how to test membership
Continuous $\rightarrow$ general equation	Cross-multiply the proportion, then collect terms	Basic algebra step glossed over in the derivation